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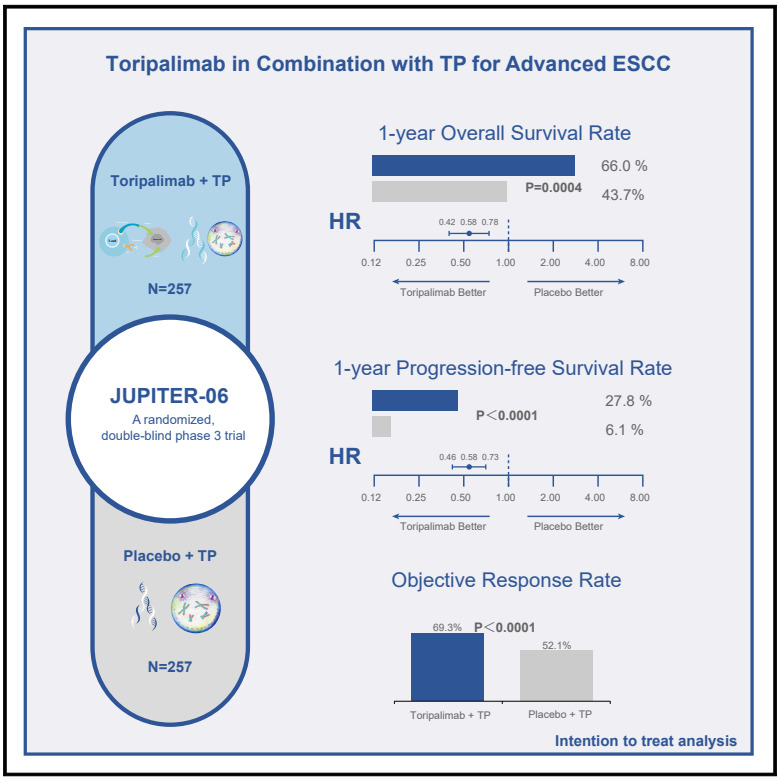
Toripalimab plus chemotherapy in treatment-naïve, advanced esophageal squamous cell carcinoma (JUPITER-06): A multi-center phase 3 trial

Graphical abstract



Highlights

- First-line toripalimab plus chemotherapy for esophageal squamous cell carcinoma
- Toripalimab plus chemotherapy improves progression-free survival and overall survival
- Toripalimab plus chemotherapy is efficacious irrespective of PD-L1 expression
- Toripalimab plus chemotherapy shows a manageable safety profile

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In brief

Wang et al. demonstrate the efficacy and safety of toripalimab plus paclitaxel/cisplatin as the first-line treatment for patients with advanced esophageal squamous cell carcinoma. As compared with paclitaxel/cisplatin alone, toripalimab plus paclitaxel/cisplatin extends progression-free survival and overall survival of the patients irrespective of PD-L1 expression.

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Toripalimab plus chemotherapy in treatment-naïve, advanced esophageal squamous cell carcinoma (JUPITER-06): A multi-center phase 3 trial

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SUMMARY

Platinum-based chemotherapy is the standard first-line treatment for advanced esophageal squamous cell carcinoma (ESCC). In this phase 3 study (ClinicalTrial.gov: NCT03829969), 514 patients with treatment-naïve advanced ESCC were randomized (1:1) to receive toripalimab or placebo in combination with paclitaxel plus cisplatin (TP) every 3 weeks for up to 6 cycles, followed by toripalimab or placebo maintenance. At the pre-specified final analysis of progression-free survival (PFS), a significant improvement in PFS is observed for the toripalimab arm over the placebo arm (hazard ratio [HR] = 0.58; 95% CI, 0.46–0.74; $p < 0.0001$). The pre-specified interim analysis of overall survival (OS) also reveals a significant OS improvement for patients treated with toripalimab plus TP over placebo plus TP (HR = 0.58; 95% CI, 0.43–0.78; $p = 0.0004$). The incidences of grade $\geq$3 treatment-emergent adverse events are similar between the two arms. Toripalimab plus TP significantly improves PFS and OS in patients with treatment-naïve, advanced ESCC, with a manageable safety profile.

INTRODUCTION

Esophageal cancer is one of the most common malignancies globally, ranking 7th in incidence and 6th in mortality among all malignancies in 2020 (Sung et al., 2021). There were over 600,000 cases of newly diagnosed esophageal cancer and over 540,000 deaths reported in 2020 worldwide (Sung et al., 2021). Histologically, esophageal cancer can be classified as esophageal squamous cell carcinoma (ESCC) or esophageal adenocarcinoma. The majority of newly diagnosed cases

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worldwide are of squamous histology. The incidence and histology vary by geographical location (Abnet et al., 2018; Qiu et al., 2021). Regions with a high incidence of esophageal cancer include Asia and Southern Africa, with over 320,000 new cases in 2020 occurring in China alone (Cao et al., 2021). ESCC is the predominant histological subtype in Asia, accounting for about 90% of all esophageal cancer cases (Abnet et al., 2018). By contrast, in the US and other Western countries, ESCC constitutes about 30% of all esophageal cancer cases (Abnet et al., 2018; Patel and Benipal, 2018).

Although surgery is the best curative treatment option for non-metastatic ESCC, postoperative relapse is common. Since the clinical symptoms of early ESCC are not distinctive, a majority of newly diagnosed ESCC patients present with advanced disease. Currently, standard first-line treatments for advanced or metastatic ESCC are combination chemotherapeutic regimens such as 5-fluorouracil or taxane plus platinum. In a retrospective, real-world evaluation, first-line chemotherapy with paclitaxel plus cisplatin (TP) appears to result in similar median progression-free survival (PFS) compared with 5-fluorouracil plus cisplatin (FP) (7.9 versus 6.5 months) in patients with advanced ESCC (Liu et al., 2016). Nevertheless, the clinical benefits of such platinum-containing doublets remain limited, with a median OS of 7 to 13 months and a 5-year overall survival (OS) rate of less than 20% (Liu et al., 2016; Petrasch et al., 1998), highlighting the urgent, unmet medical need for novel drugs and strategies for advanced ESCC.

In recent years, anti-PD-(L)1 antibodies, which are designed to interrupt immunosuppression by the PD-1/PD-L1 signaling pathway and reinvigorate T cells in the tumor microenvironment (Ribas and Wolchok, 2018), have exerted antitumor activity across multiple tumor types, including esophageal cancer (Constantinidou et al., 2019). Prior randomized studies have demonstrated significant OS improvements with anti-PD-1 therapy over standard chemotherapy for advanced esophageal cancer in the second-line or later settings (Kato et al., 2019; Kojima et al., 2020). These studies have resulted in the approvals of pembrolizumab in PD-L1 expressing, combined positive score (CPS) $\geq$10 ESCC and nivolumab in PD-L1-unselected ESCC in the second-line or later treatment settings (Kato et al., 2019; Kojima et al., 2020). Several studies have also evaluated PD-1 blockade in combination with chemotherapy in the first-line setting (Chau et al., 2021; Luo et al., 2021; Sun et al., 2021). In the KEYNOTE-590 study, pembrolizumab in combination with FP chemotherapy significantly improved OS compared with chemotherapy alone, with a median OS of 12.4 versus 9.8 months and a hazard ratio (HR) of 0.73 (95% CI, 0.62–0.86) (Sun et al., 2021). Based on these results, pembrolizumab plus FP received US Food and Drug Administration (FDA) approval for the first-line treatment of advanced esophageal cancer in March 2021.

Toripalimab, a humanized immunoglobulin G $(IgG)_{4K}$ monoclonal antibody specific for human PD-1, was approved as a monotherapy for second-line treatment of metastatic melanoma, third-line treatment of advanced nasopharyngeal carcinoma, and second-line treatment of metastatic urothelial carcinoma by the National Medical Product Administration (NMPA) in China (Sheng et al., 2020; Tang et al., 2020; Wang et al., 2021; Wei et al., 2020). Recently, the FDA has granted Breakthrough Therapy Designation to toripalimab monotherapy for patients with recurrent or metastatic non-keratinizing nasopharyngeal carcinoma with disease progression on or after platinum-containing chemotherapy (Wang et al., 2021) and to toripalimab in combination with gemcitabine and cisplatin as the first-line treatment for advanced or metastatic non-keratinizing nasopharyngeal carcinoma (Mai et al., 2021). In a prior multi-center, phase 1b/2 study, toripalimab monotherapy achieved an objective response rate (ORR) of 18.6% in 59 patients with chemotherapy refractory ESCC (Wang et al., 2019a), while toripalimab in combination with TP as a first-line treatment for advanced ESCC also showed promising efficacy with a manageable safety profile (Ren et al., 2020). In this randomized, double-blind, placebo-controlled, phase 3 study (JUPITER-06), we compared the efficacy and safety of toripalimab plus TP versus placebo plus TP as the first-line treatment for patients with advanced ESCC.

RESULTS

Patients and treatment

Between January 28, 2019 and November 30, 2020, 809 ESCC patients were screened for eligibility from 72 participating center across China, and 514 were randomly assigned 1:1 to the toripalimab plus TP arm (n = 257) or placebo plus TP arm (n = 257), stratified by the ECOG performance score (0 versus 1) and prior radiotherapy (yes versus no) (Figure 1).

Baseline demographic and disease characteristics were generally comparable between the two treatment arms, including age, gender, ECOG performance score, baseline disease stage (locally advanced not amendable for curative therapy versus metastatic), baseline tumor stage, tumor, node, metastasis (TNM) classification, site of tumor at initial diagnosis, cigarette use, alcohol use, prior radiotherapy (yes versus no), and PD-L1 expression (Tables 1 and S1). Specifically, in the toripalimab and placebo arms, PD-L1-positive expression (CPS $\geq$1) was identified in 201 (78.2%) and 200 (77.8%) patients, respectively. PD-L1-high expression, defined by CPS

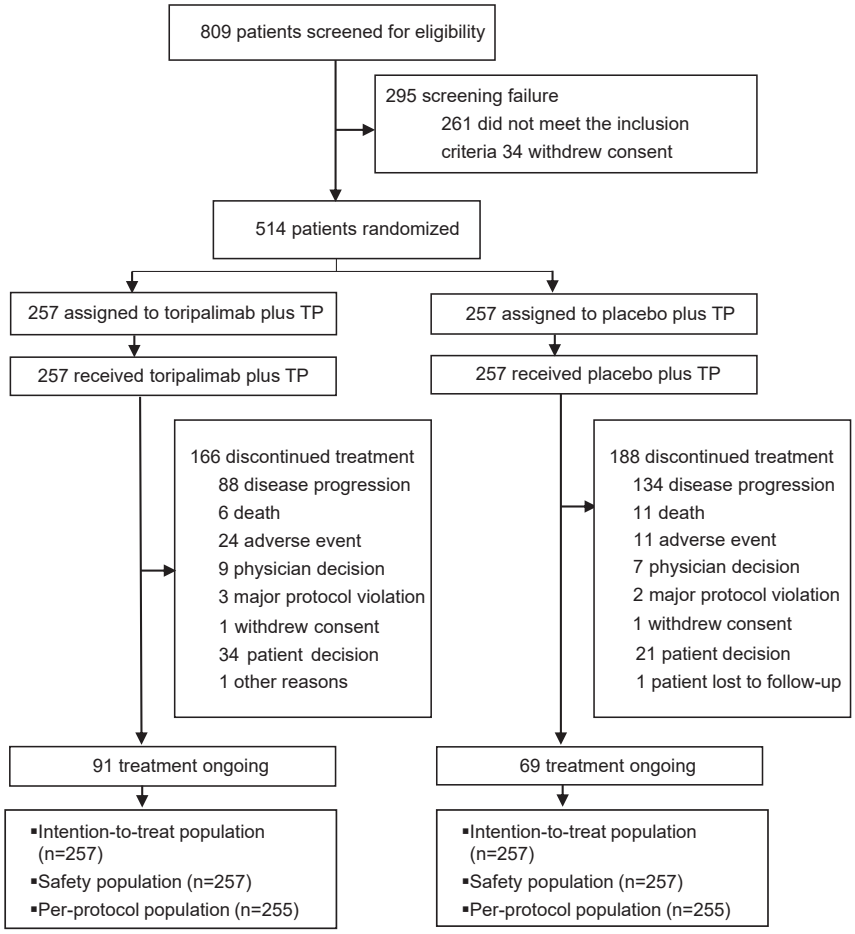


Figure 1. CONSORT diagram of the JUPITER-06 study

Between January 28, 2019 and November 30, 2020, 809 patients with advanced esophageal squamous cell carcinoma were screened for eligibility from 72 participating centers across China and 514 were randomly assigned 1:1 to the toripalimab plus chemotherapy (paclitaxel plus cisplatin [TP]) arm (n = 257) and the placebo plus TP arm (n = 257). As of March 22, 2021, the data cutoff date, all randomized patients (N = 514) received at least one dose of protocol-specified treatment and were included in the intention-to-treat analysis and safety analysis sets. Two patients from each arm were excluded from per-protocol analysis set due to major deviations for enrollment. By the cutoff date, 91 (35.4%) patients in the toripalimab arm and 69 (26.8%) in the placebo arm remained on the study treatment.

≥10, was detected in 115 (44.7%) and 97 (37.7%) patients in the two arms, respectively (Table 1).

As of March 22, 2021, the data cutoff date, all randomized patients (N = 514) received at least one dose of protocol-specified treatment and were included in the intention-to-treat (ITT) analysis and safety analysis sets (Table S2). Two patients from each arm were excluded from the per-protocol analysis set due to major protocol deviations for enrollment (Table S3). Both arms received a median of 6 cycles of paclitaxel and cisplatin and a median 7 cycles of toripalimab and 7 cycles of placebo in the two arms, respectively. By the cutoff date, 91 (35.4%) patients in the toripalimab arm and 69 (26.8%) in the placebo arm remained on the study treatment (Figure 1). The most common reason for treatment discontinuation was disease progression: 88 (34.2%) versus 134 (52.1%) in the toripalimab and placebo arms, respectively (Figure 1). For patients who discontinued the study treatment, 99/166 (59.6%) patients in the toripalimab arm and 130/188 (69.1%) in the placebo arm received subsequent anti-cancer therapy, among which 17/166 (10.2%) of the toripalimab arm and 39/188 (20.7%) of the placebo arm received anti-PD-1/PD-L1 treatment (Table S4).

PFS

By the data cutoff date of March 22, 2021, a total of 296 PFS events, as assessed by a blinded independent central review (BICR) per RECIST v.1.1, had occurred, which triggered the prespecified final PFS analysis. The median follow-up time was 7.1 months in both the toripalimab and placebo arms. There were 132 PFS events in the toripalimab arm and 164 PFS events in the placebo arm (Figure 2A; Table S5). PFS as assessed by BICR per RECIST v.1.1 was significantly better in the toripalimab chemotherapy combination arm than in the placebo chemotherapy combination arm, with a median PFS of 5.7 (95% CI, 5.6–7.0) versus 5.5 (95% CI, 5.2–5.6) months and a stratified HR for progression or death of 0.58 (95% CI, 0.46–0.74; two-sided $p < 0.0001$; Figure 2A). The 1-year PFS rates were 27.8% (95% CI, 20.4–35.8) in the toripalimab arm and 6.1% (95% CI, 2.2–12.6) in the placebo arm (Table S5). The improvement of PFS as assessed by the BICR in the toripalimab arm over the placebo arm was similar in the per-protocol analysis set (Table S6).

Treatment effects were generally more favorable in the toripalimab arm across subgroup analyses (Figure 2B). Specifically, the HRs for disease progression or death between the toripalimab and placebo arms were 0.58 (95% CI, 0.44–0.75) and 0.66 (95% CI, 0.37–1.19) in the PD-L1 CPS ≥1 and CPS <1 subgroups, respectively, and 0.65 (95% CI, 0.45–0.92) and 0.56 (95% CI, 0.41–0.78) in the CPS ≥10 and CPS <10 subgroups, respectively (Figure 2B).

Toripalimab in combination with TP chemotherapy also significantly improved the investigator-assessed PFS per RECIST v.1.1 compared with that of chemotherapy alone, with a median PFS of 7.0 (95% CI, 5.7–8.1) versus 5.6 (95% CI, 5.5–5.8) months and a stratified HR for progression or death of 0.58 (95% CI, 0.46–0.74; two-sided $p < 0.0001$). The 1-year investigator-assessed PFS rates were 27.6% (95% CI, 19.9–35.8) in the toripalimab arm and 6.1% (95% CI, 2.4–12.1) in the placebo arm (Figure S1; Table S7). The improvement in PFS as assessed by the investigator in the toripalimab arm over the placebo arm was similar in the per-protocol analysis set (Table S8).

Table 1. Summary of patient demographics

Baseline characteristic	Toripalimab + TP (n = 257)	Placebo + TP (n = 257)
Age (years)		
Median	63.0	62.0
Range	20–75	40–74
Gender, n (%)		
Male	217 (84.4)	220 (85.6)
Female	40 (15.6)	37 (14.4)
Baseline weight (kg)[a]		
Median	58.00	58.00
Range	38.5–101.0	37.0–95.0
Baseline ECOG PS, n (%)		
0	66 (25.7)	68 (26.5)
1	191 (74.3)	189 (73.5)
Cigarette use, n (%)		
Never a smoker	93 (36.2)	84 (32.7)
Former smoker	33 (12.8)	46 (17.9)
Current smoker	131 (51.0)	127 (49.4)
Alcohol use, n (%)		
Never a drinker	111 (43.2)	92 (35.8)
Former drinker	18 (7.0)	24 (9.3)
Current drinker	128 (49.8)	141 (54.9)
Baseline metastasis satus, n (%)		
Locally advanced not amendable for curative therapy	50 (19.5)	59 (23.0)
Metastatic	206 (80.2)	198 (77.0)
Unknown	1 (0.4)	0
Baseline clinical stage (%)		
I/II	3 (1.2)	4 (1.6)
III	22 (8.6)	25 (9.7)
IV	231 (89.9)	228 (88.7)
Not available	1 (0.4)	0
Site of tumor at initial diagnosis (%)		
Cervical esophagus	6 (2.3)	7 (2.7)
Upper thoracic esophagus	27 (10.5)	25 (9.7)
Middle thoracic esophagus	80 (31.1)	82 (31.9)
Lower thoracic esophagus	83 (32.3)	81 (31.5)
Esophagogastric junction	10 (3.9)	5 (1.9)
Other[b]	51 (19.8)	57 (22.2)
PD-L1 status[c], n (%)		
CPS <1	43 (16.7)	44 (17.1)
CPS ≥1	201 (78.2)	200 (77.8)
CPS <10	129 (50.2)	147 (57.2)
CPS ≥10	115 (44.7)	97 (37.7)
Unknown	13 (5.1)	13 (5.1)
Prior radiation therapy, n (%)		
Yes	35 (13.6)	35 (13.6)
No	222 (86.4)	222 (86.4)

Combined positive score (CPS) is the total number of PD-L1-staining cells (tumor cells and immune cells) divided by the total number of viable tumor cells, multiplied by 100.
[a]Baseline weight was not available from 3 patients in the placebo arm.
[b]Includes patients with tumors in multiple segments.
[c]PD-L1 assay based on JS311 immunohistochemistry staining.

OS

As the BICR-assessed PFS crossed the prespecified efficacy boundary, an interim OS analysis for efficacy was formally conducted. By the cutoff date of March 22, 2021, 173 deaths were reported: 70 (27.2%) in the toripalimab arm and 103 (40.1%) in the placebo arm in the ITT population. The interim OS analysis, which represents 47.3% of the information fraction of the final OS analysis, also crossed the prespecified boundary based on the O'Brien Fleming adjustment, demonstrating a statistically significant and clinically meaningful improvement in OS in the toripalimab arm over the placebo arm.

Among the ITT population, OS was significantly better in the toripalimab arm than in the placebo arm, with a median OS of 17 (95% CI, 14.0-not reached) versus 11 (95% CI, 10.4–12.6) months and a stratified HR for death of 0.58 (95% CI, 0.43–0.78; two-sided p = 0.0004). The 1-year OS rates were 66.0% (95% CI, 57.5–73.2) versus 43.7% (95% CI, 34.4–52.6) in the toripalimab and placebo arms, respectively (Figure 3A; Table S9). A similar OS improvement was identified in the per-protocol population (Table S10).

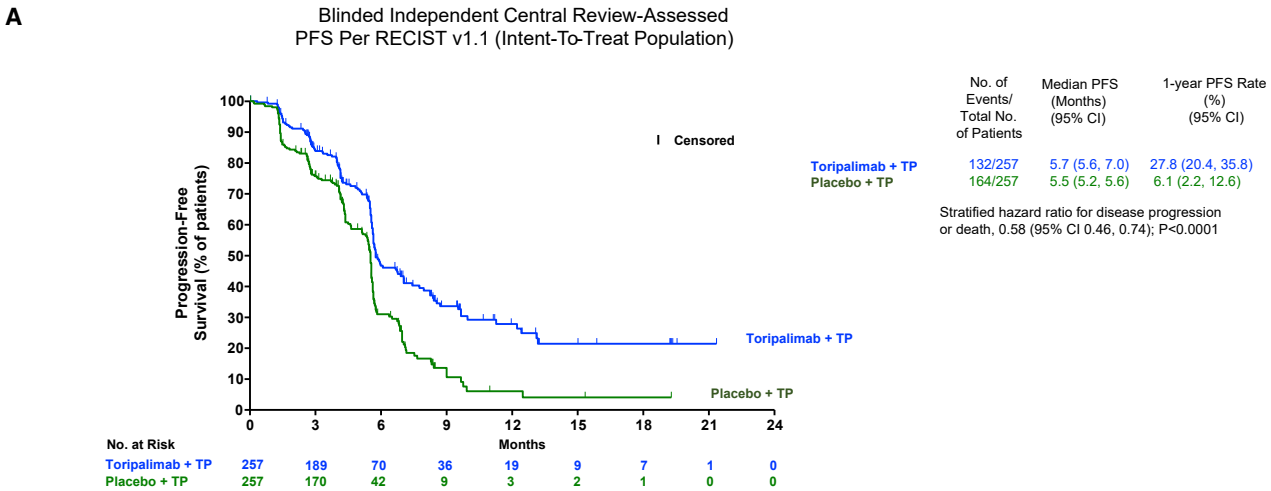
Treatment effects were generally consistent across subgroup analyses except in the female subgroup (Figure 3B). However, the insufficient number of patients (n = 77) and limited OS events (n = 22) in the female subgroup may limit a reliable assessment of treatment effects in this subgroup.

Anti-tumor response

The toripalimab chemotherapy combination arm also outperformed the placebo arm in terms of ORR, duration of response (DoR), and disease control rate (DCR) based on both BICR- and investigator-assessed results. In the ITT population, the BICR-ORR per RECIST v.1.1 was higher in the toripalimab arm than in the placebo arm: 69.3% (95% CI, 63.2–74.8) versus 52.1% (95% CI, 45.8–58.4, nominal p < 0.0001). The DCR was 89.1% (95% CI, 84.6–92.6) versus 82.1% (95% CI, 76.9–86.6, nominal p = 0.0206) (Table 2). As assessed by the BICR, the median DoR was 5.6 months (95% CI, 4.4–8.7) versus 4.2 months (95% CI, 4.2–4.4) (HR, 0.58; 95% CI, 0.41–0.81; nominal p = 0.0014) (Table 2). As assessed by the investigator per RECIST v.1.1, similar ORR, DCR, and DoR results were observed (Table S11).

Adverse events

As of March 22, 2021, the median treatment duration of exposure to toripalimab/placebo was 21 and 19.4 weeks. In the two



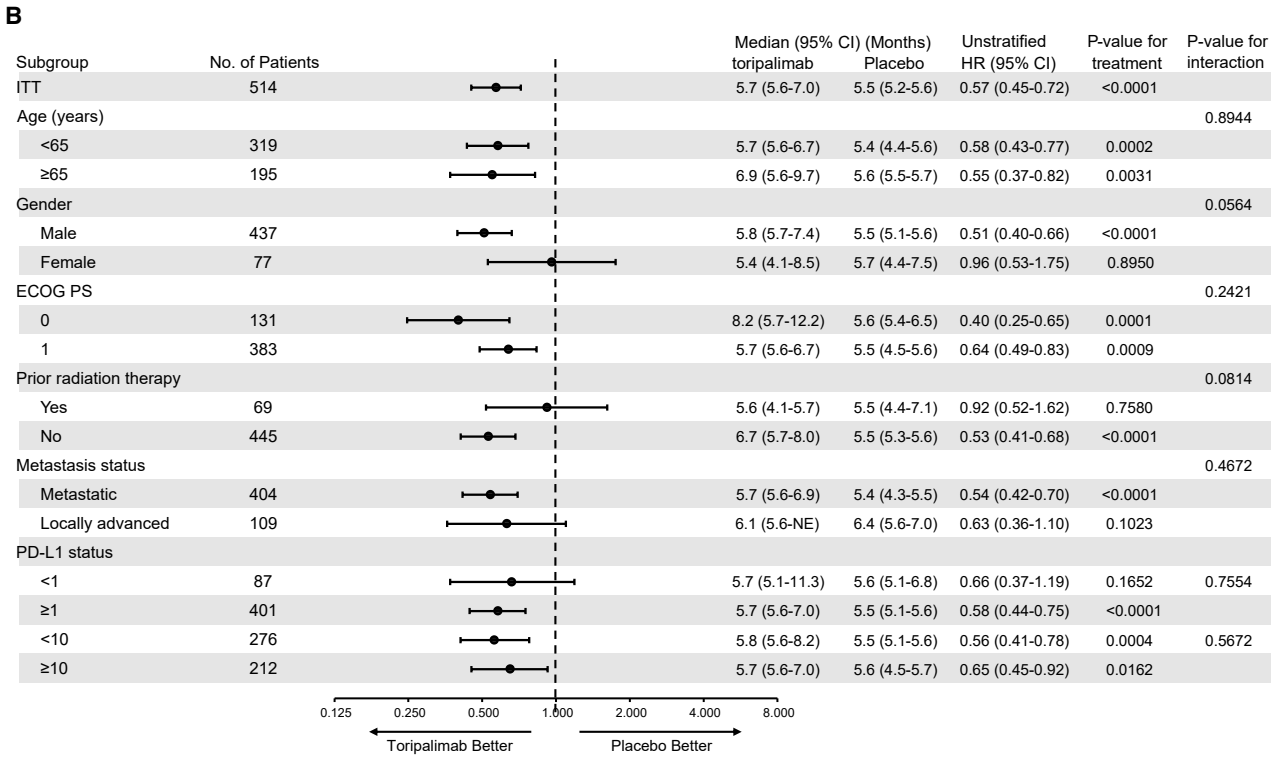


Figure 2. Progression-free survival (PFS) in the intention-to-treat population

(A) Kaplan-Meier-estimated PFS curves as assessed by the BICR according to RECIST v.1.1

Censored patients are marked with "|" in the graph. Numbers of patients at risk at indicated time points shown below the x axis. Number of events, median PFS, 1-year PFS rates, and stratified hazard ratio for PFS are shown to the right of Kaplan-Meier curves.

(B) Progression-free survival in key subgroups.

All hazard ratios were computed using the Cox proportional hazards model. All p values were two-sided with no adjustment of multiplicity. The p values of comparing the Kaplan-Meier curves were computed using the log rank test stratified by the baseline ECOG performance status (0 or 1) and prior radiotherapy (yes or no) for (A) or the unstratified log rank test for (B). The p values of testing the interaction of the subgroup variables with the treatment (B) were computed using the Cox proportional hazards regression model with the treatment arm, the subgroup variable, and their interaction as the covariates.

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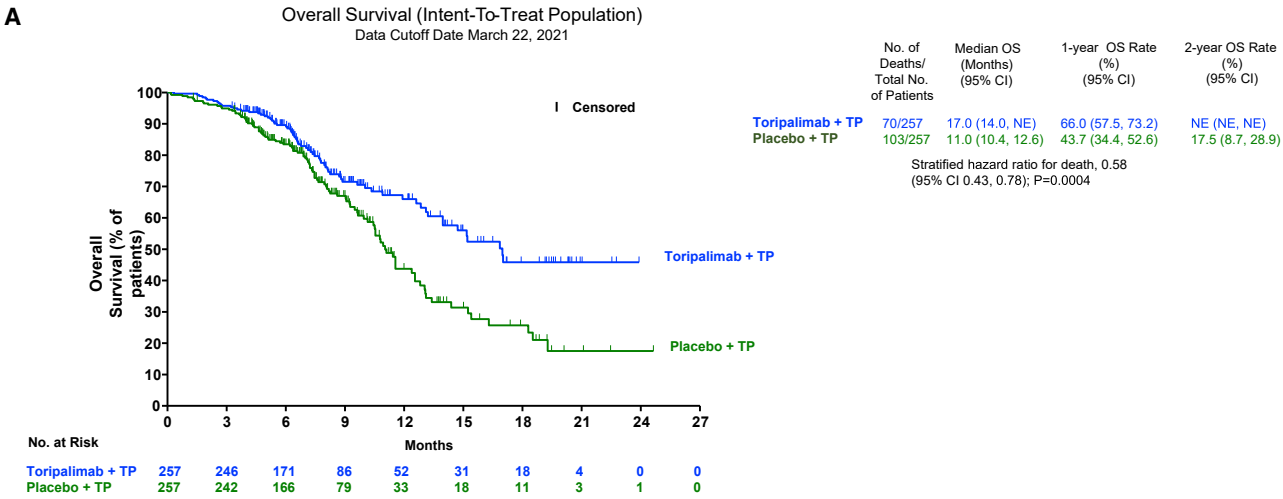

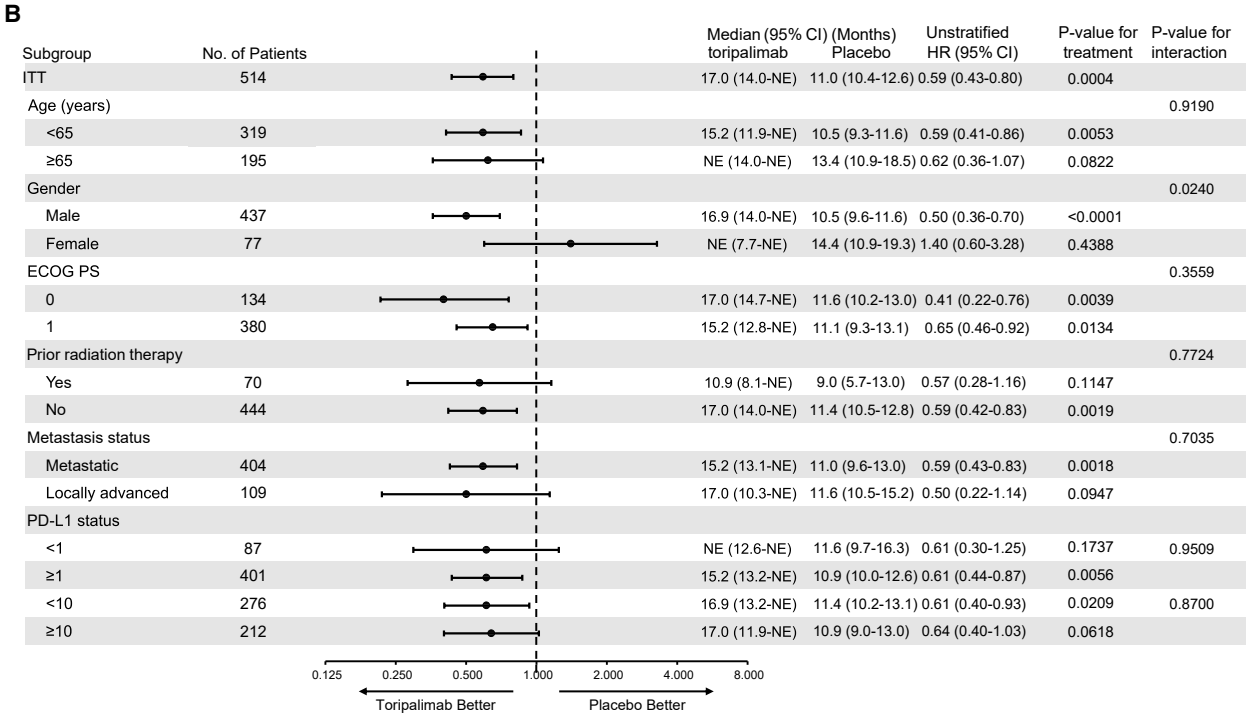

Figure 3. Overall survival (OS) in the intention-to-treat population

(A) Kaplan-Meier estimates of OS are shown to compare the toripalimab plus chemotherapy arm with the placebo plus chemotherapy arm. Censored patients are marked with "|" in the graph. Numbers of patients at risk at indicated time points are shown below the x axis. Number of events, median OS, 1-year and 2-year OS rates and stratified hazard ratio for death are shown to the right of Kaplan-Meier curves.

(B) OS in key subgroups.

All hazard ratios were computed using the Cox proportional hazards model. All p values were two-sided with no adjustment of multiplicity. The p values of comparing the Kaplan-Meier curves were computed using the log rank test stratified by the baseline ECOG performance status (0 or 1) and prior radiotherapy (yes or no) for (A) or the unstratified log rank test for (B). The p values of testing the interaction of the subgroup variables with the treatment (B) were computed using the Cox proportional hazards regression model with the treatment arm, the subgroup variable, and their interaction as the covariates.

Table 2. Tumor response with toripalimab versus placebo in combination with chemotherapy in advanced or metastatic ESCC by BICR per RECIST v.1.1

Variable	Toripalimab + TP (n = 257)	Placebo + TP (n = 257)
Best overall response, no. (%)		
Complete response	30 (11.7)	18 (7.0)
Partial response	148 (57.6)	116 (45.1)
Stable disease	51 (19.8)	77 (30.0)
Progressive disease	18 (7.0)	35 (13.6)
Non-CR/non-PD[a]	1 (0.4)	2 (0.8)
Not evaluable[b]	9 (3.5)	9 (3.5)
Objective response rate (ORR)		
ORR % (95% CI)	69.3 (63.2–74.8)	52.1 (45.8–58.4)
Difference in ORR % (95% CI)	17.2 (9.0–25.4)	
p value[c]	<0.0001	
Disease control rate (DCR)		
DCR % (95% CI)	89.1 (84.6–92.6)	82.1 (76.9–86.6)
Difference in DCR % (95% CI)	7.1 (1.1–13.1)	
p value[c]	0.0206	
Estimate of median DoR (95% CI) (months)	5.6 (4.4–8.7)	4.2 (4.2–4.4)
Stratified estimate of HR (toripalimab versus placebo) (95% CI)[d]	0.58 (0.41–0.81)	
p value[e]	0.0014	

TP, paclitaxel and cisplatin; CR, complete response; PD, progressive disease; NE, not estimable.

[a]Non-CR/non-PD: persistence of one or more non-target lesions or stable, decreasing, or mild increase in uptake of bone lesions on bone scintigraphy.

[b]Nine patients from each arm only had baseline scan and were classified as not evaluable.

[c]In the stratified analysis, nominal p value (two-sided) of the comparison between arms was calculated using the Cochran-Mantel-Haenszel method based on random stratification factors, and the 95% CI for the difference in the rate difference between the arms was estimated based on Mantel-Haenszel, stratified by ECOG PS score (0 versus 1), and prior radiotherapy (yes versus no).

[d]HR for DoR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were ECOG PS score (0 versus 1) and prior radiotherapy (yes versus no).

[e]p value (two-sided) for DoR was calculated using log rank test, stratified analysis was performed using stratified log rank test, and the stratification factors were the same as HR.

arms, the median exposures for cisplatin were 15.1 and 15 weeks, and the median exposures for paclitaxel were 15.2 and 15 weeks (Table S12). Taken together, these results indicate that concomitant toripalimab administration did not affect the ability to deliver planned chemotherapy.

By the cutoff date, the majority of patients (255, 99.2%) in both arms experienced at least 1 treatment-emergent adverse event (TEAE) (Table S13). Grade 3 and above TEAEs occurred in 188 (73.2%) patients in the toripalimab arm and 180 (70.0%) patients in the placebo arm. Fatal TEAEs were reported in 21 (8.2%) patients in each arm, while fatal adverse events that were related to study treatment (cisplatin/paclitaxel/toripalimab/placebo) occurred in 1 (0.4%) versus 3 (1.2%) patients in the toripalimab and placebo arms, respectively. Similar incidences of serious adverse events (36.2% versus 28.8%) and infusion-related reaction (3.5% versus 3.1%) were reported in the two arms. TEAEs leading to the discontinuation of toripalimab or placebo occurred in 30 (11.7%) and 16 (6.2%) patients, respectively (Table S13).

TEAEs reported in at least 10% of the patients in either arm are listed in Table 3. The most common TEAEs in the toripalimab and placebo arms included anemia (78.2% versus 80.5%), leukopenia (67.7% versus 53.7%), neutropenia (67.3% versus 54.1%), nausea (43.2% versus 45.9%), fatigue (42.8% versus 39.3%), neuropathy peripheral (40.5% versus 43.2%), vomiting (40.1% versus 36.6%), decreased appetite (39.3% versus 45.9%), and alopecia (35.4% versus 40.5%). There were modest differences in the incidence of thrombocytopenia (28.8% versus 16.7%), diarrhea (23% versus 14%), rash (21.8% versus 10.5%), and pruritis (11.7% versus 2.7%) in the toripalimab arm compared with the placebo arm. The most common grade 3 or above TEAEs in the toripalimab and placebo arms included neutropenia (42.4% versus 33.1%), leukopenia (20.2% versus 13.2%), and anemia (10.9% versus 14.8%) (Table 3).

Immune-related adverse events (irAEs), as assessed by the investigator while blinded to study treatment, were more frequent in the toripalimab arm than the placebo arm (37.0% versus 26.5%), as were the grade 3 and above irAEs (7% versus 1.6%) (Table S13). The most common irAEs, with an incidence $\geq$4% in the toripalimab arm, included rash (10.1% versus 3.5%), hypothyroidism (5.8% versus 3.5%), and pruritus (4.3% versus 1.6%) (Table S14). Three patients experienced immune-mediated hepatitis, including 1 patient (grade 4) from the toripalimab arm and 2 patients (grades 3 and 1) from the placebo arm. In addition, one patient from the toripalimab arm experienced immune-mediated myocarditis (grade 3) (Table S14).

DISCUSSION

In this double-blind, randomized phase 3 study, JUPITER-06, we compared the efficacy and safety of toripalimab in combination with TP versus placebo with TP for the first-line treatment of locally advanced or metastatic ESCC. At the prespecified final PFS and interim OS analyses, our results demonstrated that the addition of toripalimab to the standard first-line chemotherapy resulted in statistically significant and clinically meaningful improvement in both PFS and OS than chemotherapy alone. Moreover, toripalimab plus TP showed a manageable tolerability profile. As the p value of interim OS analysis has crossed the prespecified efficacy boundary, the interim OS result is thus considered definitive. The planned final OS analysis will be for information only.

Although the PFS curves were relatively close at the 50% PFS probability (median PFS, 5.7 versus 5.5 months for the toripalimab versus placebo arms), the curves separate early, and the curve of the toripalimab arm is above that of the placebo arm over the entire period, resulting in a PFS HR of 0.58 and a

Table 3. Treatment-emergent adverse events reported in at least 10% of the patients in the two treatment arms

	Toripalimab + TP (n = 257) no. (%)		Placebo + TP (n = 257) no. (%)	
Adverse event, no. of patients (%)	All grades	grade ≥3	all grades	grade ≥3
Any TEAE	255 (99.2)	188 (73.2)	255 (99.2)	180 (70.0)
Anemia	201 (78.2)	28 (10.9)	207 (80.5)	38 (14.8)
Leukopenia	174 (67.7)	52 (20.2)	138 (53.7)	34 (13.2)
Neutropenia	173 (67.3)	109 (42.4)	139 (54.1)	85 (33.1)
Nausea	111 (43.2)	0 (0)	118 (45.9)	5 (1.9)
Fatigue	110 (42.8)	11 (4.3)	101 (39.3)	5 (1.9)
Neuropathy peripheral	104 (40.5)	3 (1.2)	111 (43.2)	10 (3.9)
Vomiting	103 (40.1)	5 (1.9)	94 (36.6)	5 (1.9)
Decreased appetite	101 (39.3)	3 (1.2)	118 (45.9)	5 (1.9)
Alopecia	91 (35.4)	0 (0)	104 (40.5)	0 (0)
Weight decreased	74 (28.8)	8 (3.1)	73 (28.4)	5 (1.9)
Thrombocytopenia	74 (28.8)	5 (1.9)	43 (16.7)	3 (1.2)
Hypoproteinemia	63 (24.5)	0 (0)	61 (23.7)	0 (0)
Constipation	62 (24.1)	0 (0)	55 (21.4)	0 (0)
Musculoskeletal pain	61 (23.7)	3 (1.2)	52 (20.2)	4 (1.6)
Diarrhea	59 (23.0)	1 (0.4)	36 (14.0)	1 (0.4)
Cough	56 (21.8)	0 (0)	52 (20.2)	1 (0.4)
Rash	56 (21.8)	8 (3.1)	27 (10.5)	1 (0.4)
Hypokalemia	44 (17.1)	8 (3.1)	49 (19.1)	17 (6.6)
Pyrexia	41 (16.0)	3 (1.2)	28 (10.9)	0 (0)
Hyponatremia	38 (14.8)	9 (3.5)	47 (18.3)	4 (1.6)
Blood creatinine increased	37 (14.4)	1 (0.4)	29 (11.3)	0 (0)
Abdominal pain	36 (14.0)	2 (0.8)	29 (11.3)	0 (0)
Hypomagnesemia	31 (12.1)	2 (0.8)	35 (13.6)	8 (3.1)
Hyperglycemia	30 (11.7)	1 (0.4)	25 (9.7)	0 (0)
Aspartate aminotransferase increased	30 (11.7)	1 (0.4)	23 (8.9)	0 (0)
Pruritus	30 (11.7)	1 (0.4)	7 (2.7)	0 (0)
Insomnia	29 (11.3)	0 (0)	26 (10.1)	0 (0)
Hyperlipidemia	27 (10.5)	1 (0.4)	28 (10.9)	2 (0.8)
Alanine aminotransferase increased	27 (10.5)	1 (0.4)	24 (9.3)	1 (0.4)
Pneumonia	26 (10.1)	15 (5.8)	20 (7.8)	12 (4.7)
Hyperuricemia	24 (9.3)	1 (0.4)	31 (12.1)	0 (0)
Dysphagia	23 (8.9)	6 (2.3)	33 (12.8)	8 (3.1)

TEAE, treatment-emergent adverse event. Adverse event coding version: MedDRA 24.0. Adverse events that occurred during treatment are all adverse events that occurred between the first dose and 60 days after the last dose or before starting a new antineoplastic therapy, whichever occurred first. For the same patient, if the same adverse event occurred multiple times, that patient is counted only once in the same system organ classification and preferred terminology when calculating the number of adverse events. Data cutoff: March 22, 2021.

clinically significant improvement in the 1-year PFS rate (27.8% versus 6.1%).

There have been previous efforts in exploring the role of PD-1 blockade plus chemotherapy in first-line treatment of advanced ESCC. The phase 3 KEYNOTE-590 study demonstrated significant PFS and OS improvements with the addition of pembrolizumab to FP as first-line therapy for patients with advanced ESCC (Sun et al., 2021). Additionally, the phase 3 CheckMate 648 study, which predominantly enrolled patients with advanced ESCC, showed that nivolumab plus FP significantly prolonged PFS and OS compared with chemotherapy alone as a first-line treatment for the PD-L1-postive population as well as the ITT population (Chau et al., 2021). Similar to the findings from KEYNOTE-590 and CheckMate 648, the current JUPITER-06 study showed the superiority of toripalimab plus TP over placebo plus TP in terms of PFS and OS. TP was chosen as the chemotherapy backbone in this study according to both international and local treatment guidelines (Ajani et al., 2019; Muro et al., 2019; National Health Commission of the People's Republic of China, 2019). The combination of TP with another PD-1 antibody, camrelizumab, in advanced ESCC also showed significant improvements in PFS and OS (Luo et al., 2021).

Previous studies have shown that PD-L1 expression on tumor cells or tumor-infiltrating lymphocytes correlated with favorable responses to anti-PD-1/PD-L1 therapy in certain solid tumors

(Brahmer et al., 2015; Rizvi et al., 2015; Wang et al., 2019b). In esophageal cancer, KENOTE-590, CheckMate 648, and ESCORT-1st all suggest a positive correlation between the PD-L1 expression level (either on tumor cells or both tumor and immune cells) and the efficacy of PD-1 blockade plus chemotherapy (Chau et al., 2021; Luo et al., 2021; Sun et al., 2021). However, the current study showed that the PFS and OS benefits of the the toripalimab-TP combination were independent of the PD-L1 expression level. In prior studies performed among different PD-L1 immunohistochemistry (IHC) assays, JS311 has shown similar PD-L1 staining results with the 22C3, 28-8, and SP263 antibodies in multiple cancer types including non-small cell lung cancer, urothelial cancer, and melanoma (Sheng et al., 2021; Wang et al., 2020), as well as in ESCC (Table S15). Similar to SP263, JS311 also binds to an epitope in the cytoplasmic domain of PD-L1, whereas 22C3 and 28-8 recognize the extracellular domain of PD-L1 (Lawson et al., 2020). Therefore, we may speculate that the lack of correlation between treatment effects and different PD-L1 expression levels is unlikely explained by the usage of a different PD-L1 staining assay in JUPITER-06. The reasons for the differences in the correlation of PD-L1 expression and efficacy between JUPITER-06 and the previous phase 3 trials remain unclear. However, similar to KENOTE-590, CheckMate 648, and ESCORT-1st, the JUPITER-06 study also demonstrates statistically significant PFS and OS benefits of the toripalimab-TP combination among the ITT population, hence supporting the use of this regimen in advanced ESCC irrespective of PD-L1 expression.

Consistent with the previous studies investigating PD-1 blockade plus chemotherapy in advanced ESCC (Chau et al., 2021; Luo et al., 2021; Sun et al., 2021), the current study showed that the addition of toripalimab to first-line TP did not lead to an unacceptable increase in toxicity. The incidence of grade 3 or above TEAEs, serious adverse events, and fatal TEAEs were comparable between the two arms. Moreover, the toxicity profile observed in JUPITER-06 was consistent with those previously observed in other PD-1 blockers in combination with chemotherapy (Luo et al., 2021), as well as those in toripalimab plus chemotherapy in gastric cancer (Wang et al., 2019b) and nasopharyngeal carcinoma (Mai et al., 2021), with no new safety signal identified.

In June 2020, pembrolizumab and camrelizumab were approved for the second-line treatment of ESCC in China. However, as neither of these agents was covered by national insurance until March 2021, PD-1 inhibitors have not been effectively available to the majority of ESCC patients in the second-line setting during the conduction of the JUPITER-06 study. As a result, only 15.2% (39/257) of patients from the placebo arm received subsequent anti-PD-1/PD-L1 treatment. Consistently, in the ESCORT-1st study conducted in similar time window in China, only 13.4% (40/298) of patients in the placebo group received subsequent anti-PD-1/PD-L1 therapy.

Limitations

All patients enrolled in the JUIPTER-06 were Chinese with 100% squamous histology, which is the predominant subtype in Asia but accounts for only 30% of esophageal cancer in the West (Abnet et al., 2018; Patel and Benipal, 2018); therefore, the efficacy of the toripalimab-TP combination in esophageal adenocarcinoma and/or in the white population needs to be further explored. However, we speculate that the findings from JUPITER-06 might be extrapolated to western ESCC patients, as the efficacy and survival of the immuno-chemotherapy combination did not differ by race in the CheckMate 648 study, which enrolled both white and Asian ESCC patients (Chau et al., 2021). In addition, the current treatment guidelines regarding the management of advanced ESCC are similar in the West and East (Ajani et al., 2019; Muro et al., 2019; National Health Commission of the People's Republic of China, 2019).

Conclusions

JUPITER-06 demonstrated significant PFS and OS improvements with the addition of toripalimab to the TP chemotherapy, with an acceptable safety profile in patients with treatment-naïve advanced ESCC. Long-term OS results will be continuously collected. Our study provides a new front-line treatment option for managing this challenging disease.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.ccell.2022.02.007.

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AUTHOR CONTRIBUTIONS

Conception and design, R.-H.X., F.W., H.F., and S. Yao; provision of study materials or patients, F.W., Z.-X.W., C.C., J.Y., Yanqiao Zhang, M.L., J.F.,

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DECLARATION OF INTERESTS

X.L. is employed by Shanghai Junshi Biosciences. X.T., H.F., and S. Yao are employed by Shanghai Junshi Biosciences and TopAlliance Biosciences. R.M. and P.K. are employed by TopAlliance Biosciences. R.-H.X. has served as a consulting or advisory role for Bristol-Myers Squibb, Merck Serono, Roche, Astellas, and AstraZeneca. All other authors have no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Toripalimab	Shanghai Junshi Biosciences	N/A
Biological samples		
Tissue biopsy samples	Participating study centers	N/A
Chemicals, peptides, and recombinant proteins		
Paclitaxel	Hospira Australia Pty Ltd and Bristol Myers Squibb	N/A
Cisplatin	Qilu Pharmaceutical Co. Ltd.	N/A
Critical commercial assays		
JS311 antibody for PD-L1 immunohistochemistry staining	MEDx Translational Medicine Co. Ltd.	P003B01.a
Software and algorithms		
SAS version 9.3	SAS Institute	RRID: SCR_008567

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Feng Wang (wangfeng@sysucc.org.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

All requests for data will be reviewed by the leading clinical site Sun Yat-Sen University Cancer Center and the study sponsor, Shanghai Junshi Biosciences, to verify if the request is subject to any intellectual property or confidentiality obligations. Requests for access to the patient-level data from this study can be submitted via email to xurh@sysucc.org.cn with a detailed proposal for approval. A signed data access agreement with the sponsor is required before accessing the shared data. No custom code was used for statistical analysis for this study.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Study design and participants

JUPITER-06 was a multi-center, randomized, double-blind, placebo-controlled, phase 3 study conducted in 72 medical centers across China. Eligible patients were aged 18-75 years old, had histologically or cytologically confirmed, locally advanced, recurrent, or metastatic ESCC that was not amendable for curative esophagectomy or definitive chemoradiation were enrolled. Patients must have received either no prior systemic antitumor therapy for relapsed or metastatic disease or received (neo)adjuvant chemo(radio) therapy for non-metastatic disease and experienced relapse of disease following at least 6 months after the last dose of chemotherapy (an interval of at least 12 months was required after the last dose if the prior (neo)adjuvant therapy utilized the TP regimen). In addition, enrolled patients must have had an ECOG performance status of 0 or 1, a life expectancy of at least 3 months, adequate organ function, at least one measurable lesion in accordance with RECIST v1.1, and had agreed to provide either an archival or fresh tumor specimen for biomarker analysis. Key exclusion criteria included the presence of malignant tumors except esophageal cancer within 5 years prior to randomization; active or untreated CNS metastasis; active or previous autoimmune or inflammatory disorders; uncontrolled, concurrent illness or active infections; previous use of immune checkpoint blockade, systemic immune-stimulators, and/or systemic immunosuppressive drugs.

Study oversight

This randomized, double-blind, placebo-controlled trial enrolled patients from 72 sites in China. The trial protocol and amendments were approved by an institutional review board or ethics committee at each participating site. All patients provided written informed consent before enrollment. This trial was conducted in full conformance with the ICH E6 guideline for Good Clinical Practice (GCP) and the principles of the Declaration of Helsinki. The iDMC periodically assessed the safety data and reviewed the efficacy results at the interim/final analysis of PFS. All analyses for iDMC's review were prepared by an independent third party.

METHOD DETAILS

Randomization and masking

Permuted block with flexible block size (4 or 6) was used to generate the randomization allocation sequence. Randomization was stratified by baseline ECOG performance status (0 or 1) and prior receipt of radiotherapy (yes vs. no). A new block was assigned to a stratum when the first patient in that stratum was being randomized or all the randomization allocation sequences in the block assigned to that stratum had been used. Within one block, the randomization allocation sequence was assigned to patients in the fixed order as generated.

The random allocation sequences for patients and investigational drugs were generated and maintained by an independent unblinded statistician from a third-party vendor; all other personnel on the study were blinded including the study personnel, the investigator, study center staff, patient, sponsor, clinical contract research organization and the independent review committee. The random allocation sequence for investigational drugs was also held by the Clinical Supplies Department of the Sponsor, for the purpose of drug packaging. The patient randomization and the dispense of investigational drugs were implemented via the Interactive Web Response System (IWRS) – Balance (Medidata Solutions). Identically packaged toripalimab/placebo-saline infusion solutions were prepared and supplied to blinded clinical site staff for administration at scheduled infusion visit. The preparation and administration of TP chemotherapy were open label.

Procedures

Patients were randomly assigned 1:1 to receive either toripalimab or placebo in combination with TP once every 3 weeks for up to 6 cycles, followed by either toripalimab or placebo monotherapy. All medicines were given through intravenous infusion. In the chemotherapy phase, patients would receive toripalimab (240 mg) or placebo on Day 1, paclitaxel (175 mg/m^2) and cisplatin (75 mg/m^2) on Day 1 of each 3-week cycle. Chemotherapy was to continue until progressive disease, intolerable toxicity, withdrawal of consent, or a maximum of 6 cycles, whichever occurred first. In the maintenance phase, patients would receive toripalimab (240 mg) or placebo every 3 weeks until progressive disease, intolerable toxicity, withdrawal of consent or a maximum of 2 years of treatment. Crossover was not permitted as part of the study.

Tumor assessments were performed at baseline, every 6 weeks in the first 12 months and then every 9 weeks thereafter until disease progression, loss of clinical benefit, withdrawal of consent, initiation of new anti-cancer therapy or death, whichever occurred first. Clinical response was assessed by the investigator and BICR per RECIST v1.1 and the Immune-related Response Evaluation Criteria in Solid Tumors (irRECIST).

Safety and tolerability were assessed in all patients who received any amount of the study drug up to 60 days after the last dose of the study treatment, or the start of a new antitumor therapy, whichever occurred first. All adverse events were defined and graded according to the National Cancer Institute-Common Terminology Criteria for Adverse Events (NCI-CTCAE) version 5.0.

Archival or fresh tumor biopsy samples were obtained from patients prior to treatment. PD-L1 expression was evaluated by immunohistochemistry (IHC) staining with the JS311 antibody using a validated staining assay on the Ventana Benchmark Ultra platform in a central lab. JS311 is recombinant monoclonal rabbit anti-human PD-L1 antibody developed for IHC staining (Wang et al., 2020). JS311 was generated by immunization of rabbits with the cytoplasmic domain of PD-L1. A cross-correlation study performed among different PD-L1 IHC assays and JS311 showed similar PD-L1 staining patterns and scores with the 22C3, 28-8 and SP263 antibodies (Sheng et al., 2021; Wang et al., 2020). In ESCC, JS311 (CPS $\geq$ 10) showed 84% and 90% overall concordance with 22C3 (CPS $\geq$ 10) and 28-8 (TPS $\geq$ 1) for PD-L1 IHC staining (Table S15). PD-L1 CPS is defined as the total number of PD-L1 staining cells (tumor cells and immune cells) divided by the total number of viable tumor cells, multiplied by 100. PD-L1 positivity was defined as CPS $\geq$ 1.

Endpoints

The co-primary endpoints were PFS as assessed by BICR per RECIST v1.1 in the intention-to-treat (ITT) population and OS in the ITT population. Secondary endpoints included PFS assessed by the investigator, PFS in the per-protocol population, ORR, duration of response (DoR), disease control rate (DCR), PFS and OS rates at 1-year and 2-year, and safety. PFS was defined as the time from randomization to the time of the first documented disease progression or death from any cause. OS was defined as the time from randomization to the time of death from any cause. ORR was defined as the proportion of patients with confirmed complete or partial response. DoR was defined as the time between the first documentation of a response and the first documented disease progression or death. DCR was defined as the proportion of patients with the best response of complete response, partial response or stable disease.

QUANTIFICATION AND STATISTICAL ANALYSIS

The sample size of the study was determined based on the requirement of detecting a significant result for the co-primary endpoints of PFS and OS based on the following assumptions: A total of 500 patients were needed to observe 283 PFS events at approximately 24 months after the first subject was randomized in order to detect the PFS improvement of hazard ratio (HR) = 0.7 (median PFS for the control arm presumed to be 5 months) with 85% power at an overall two-sided significance level of 0.05. In addition, a total of 500 patients were needed to observe 366 OS events at approximately 39 months after the first subject was randomized in order to

detect the OS improvement of HR = 0.73 (median OS for the control arm presumed to be 10 months) with 85% power at an overall two-sided significance level of 0.05.

A hierarchical testing procedure was implemented to control for the type-1 error rate. If the primary endpoint for PFS was met, the alpha was recycled for the OS analysis. There was no alpha adjustment for testing of secondary endpoints.

The study was designed to have an interim OS analysis performed at the time of the final PFS analysis, with an expectation that 212 OS events will have occurred at the time of the final PFS analysis. However, the exact numbers of OS events at the interim analysis was determined by the timing of the PFS analysis. To control for the overall type I error for the interim and final analyses of OS at a two-sided significance level of 0.05, the stopping boundaries for the OS interim and final analyses were computed using the Lan-DeMets alpha spending function to approximate the O'Brien Fleming boundary (two-sided P=0.0022 at the interim OS analysis with 173 events; two-sided P=0.0493 at final OS analysis with 366 events). Accordingly, the interim OS analysis will be considered statistically significant if the P value in the interim OS analysis is less than 0.0022, whereas the final OS analysis will be considered statistically significant if the P value in the final OS analysis is less than 0.0493 and the interim OS analysis is not significant.

Efficacy was analyzed among the ITT population, which included all the patients that underwent randomization. Sensitivity analyses were performed for PFS (by BICR per RECIST v1.1) and OS in the per-protocol population, which excluded patients with major protocol violations. The two-sided log-rank test, stratified by ECOG performance status (0 vs. 1), and prior radiotherapy (yes vs. no), as recorded in the IWRS, was used as the primary analysis to compare PFS between the two treatment arms. The HR for disease progression or death was estimated with the use of the stratified Cox proportional hazards model. Exact method was used to handle ties. The 95% CI for the HR was constructed. The Kaplan-Meier methodology was applied to estimate the median PFS for each treatment arm. The Brookmeyer Crowley methodology, by using log-log transformation to survival function, was used to construct the 95% CI for the median PFS for each treatment arm. The methods outlined for PFS were also used for the OS and DoR analysis. A test for interaction (P interaction) was conducted using Cox proportional hazards regression models (with treatment group, subgroup variable, interaction of treatment and subgroup as covariates) to evaluate subgroup differences.

Treatment differences in ORR and DCR were analyzed using the stratified Cochran-Mantel-Haenszel test. The 95% CI for the difference in ORRs between the two treatment arms was determined using the normal approximation to the binomial distribution.

Safety analyses were performed on the safety population, which included all randomized patients who received any amount of the study drug. Serious adverse events (SAEs) leading to death, and adverse events leading to permanent discontinuation of the study drug were tabulated for each arm.

On March 22, 2021, the number of events for disease progression or death triggered the PFS analysis. The efficacy and safety results were evaluated by the independent data monitoring committee (iDMC) on April 30, 2021. The iDMC reported that the efficacy boundary for both PFS and OS had been crossed. All safety and efficacy results reported here are based on the final PFS and interim OS analyses as of Mach 22, 2021.

The data in this study were analyzed using SAS version 9.3 by SAS Institute. This trial is registered with ClinicalTrials.gov, NCT03829969.

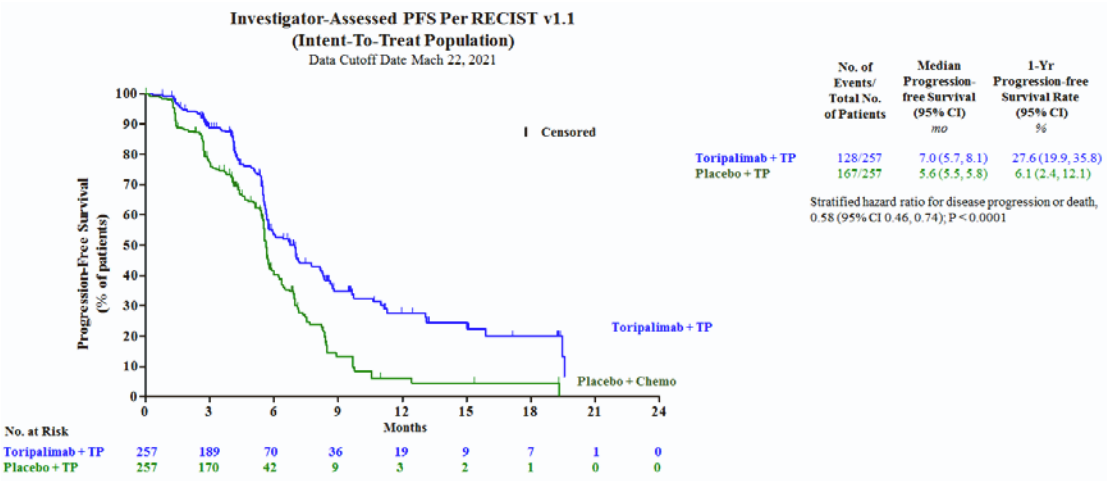
Cancer Cell, Volume *40*

Supplemental information

Toripalimab plus chemotherapy in treatment-naïve, advanced esophageal squamous cell carcinoma (JUPITER-06): A multi-center phase 3 trial

Zi-Xian Wang, Chengxu Cui, Jun Yao, Yanqiao Zhang, Mengxia Li, Jifeng Feng, Shujun Yang, Yun Fan, Jianhua Shi, Xizhi Zhang, Lin Shen, Yongqian Shu, Cailian Wang, Tianyang Dai, Teng Mao, Long Chen, Zengqing Guo, Bo Liu, Hongming Pan, Shundong Cang, Yi Jiang, Junye Wang, Min Ye, Zhendong Chen, Da Jiang, Qin Lin, Wei Ren, Junsheng Wang, Lin Wu, Yong Xu, Zhanhui Miao, Meili Sun, Conghua Xie, Ying Liu, Qifeng Wang, Lina Zhao, Qi Li, Canhong Huang, Ke Jiang, Kunyu Yang, Daojun Li, Yunpeng Liu, Zhitu Zhu, Rixin Chen, Liqun Jia, Wei Li, Wangjun Liao, Hong-Xu Liu, Daiyuan Ma, Jie Ma, Yanru Qin, Zhihong Shi, Qichun Wei, Ke Xiao, Yan Zhang, Ying Zhang, Xin Chen, Guanghai Dai, Jianxing He, Junhe Li, Guanghui Li, Yong Liu, Zhihua Liu, Xianglin Yuan, Junping Zhang, Zhichao Fu, Yifu He, Fang Ju, Zheng Liu, Peng Tang, Tiejun Wang, Weibo Wang, Jing Zhang, Xianming Luo, Xiongwen Tang, Rena May, Hui Feng, Sheng Yao, Patricia Keegan, Rui-Hua Xu, and Feng Wang

Supplementary Figure 1. Kaplan-Meier-estimated PFS curves as assessed by the investigator according to RECIST v1.1. Censored patients are marked with " ┃ " in the graph. Numbers of patients at risk at indicated time points shown below x-axis. Number of events, median PFS, 1-year PFS rates and stratified hazard ratio for PFS shown to the right of Kaplan-Meier curves (Related to Figure 2).



Supplemental Table 1. TNM classification at enrollment (Related to Table 1).

	Toripalimab + TP (N=257)	Placebo + TP (N=257)	Total (N=514)
TNM Grade primary tumor (T)			
Tx	100 (38.9)	79 (30.7)	179 (34.8)
T0	12 (4.7)	16 (6.2)	28 (5.4)
T1a/T1b	3 (1.2)	4 (1.6)	7 (1.4)
T2	14 (5.4)	18 (7.0)	32 (6.2)
T3	79 (30.7)	86 (33.5)	165 (32.1)
T4a/T4b	47 (18.3)	46 (17.9)	93 (18.1)
Not available	2 (0.8)	8 (3.1)	10 (1.9)
TNM Grade regional lymph node (N)			
Nx	82 (31.9)	59 (23.0)	141 (27.4)
N0	15 (5.8)	10 (3.9)	25 (4.9)
N1	49 (19.1)	53 (20.6)	102 (19.8)
N2	59 (23.0)	70 (27.2)	129 (25.1)
N3	51 (19.8)	61 (23.7)	112 (21.8)
Not available	1 (0.4)	4 (1.6)	5 (1.0)
TNM Grade Metastasis (M)			
M0	51 (19.8)	59 (23.0)	110 (21.4)
M1	206 (80.2)	198 (77.0)	404 (78.6)

Supplemental Table 2. Analysis data sets (Related to Table 1).

	Toripalimab + TP (N=257)	Placebo + TP (N=257)	Total (N=514)
Randomized	257	257	514
Intention-to-treat Analysis Set	257	257	514
Per-Protocol Analysis Set	255 (99.2%)	255 (99.2%)	510 (99.2%)
Safety Analysis Set	257 (100%)	257 (100%)	514 (100%)

Supplemental Table 3. Per-Protocol Analysis Set Exclusions (Related to Table 1).

Subject ID / Arm	Major Protocol Violation
1007011 / toripalimab	Prior systemic anti-tumor therapy (treatment regimen contained docetaxel and nedaplatin), which did not meet the inclusion criterion.
1009002 / toripalimab	A second primary gastric adenocarcinoma, which met an exclusion criterion.
1015014 / Placebo	Identified to have hypopharyngeal carcinoma in situ by pathology report after enrollment, which met an exclusion criterion.
1047002 / Placebo	Uncontrolled tumor related pain before enrollment, which met an exclusion criterion.

Supplemental Table 4. Subsequent Anti-Cancer Therapy in Intent-to-Treat Population After the Study Treatment (Related to Table 1).

Treatment	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)	Total (N=514) n (%)
Patients with at least one subsequent anti-cancer therapy	99 (38.5)	130 (50.6)	229 (44.6)
Chemotherapy	62 (24.1)	76 (29.6)	138 (26.8)
Radiation therapy	30 (11.7)	44 (17.1)	74 (14.4)
Targeting therapy	23 (8.9)	46 (17.9)	69 (13.4)
Anti-PD-1/PD-L1 therapy[1]	17 (6.6)	39 (15.2)	56 (10.9)
Camrelizumab	10 (3.9)	16 (6.2)	26 (5.1)
Toripalimab	4 (1.6)	10 (3.9)	14 (2.7)
Sintilimab	3 (1.2)	9 (3.5)	12 (2.3)
Pembrolizumab	0	3 (1.2)	3 (0.6)
Tislelizumab	0	1 (0.4)	1 (0.2)
Other anti-PD-1	1 (0.4)	0	1 (0.2)
Surgery	4 (1.6)	4 (1.6)	8 (1.6)
Others[2]	26 (10.1)	31 (12.1)	57 (11.1)

[1] One patient from the toripalimab arm received two anti-PD-1therapies after discontinuation from the study treatment.

[2] Others: includes traditional oriental herbal therapy, photodynamic therapy and other unclassified anti-tumor therapy.

Supplemental Table 5. Progression-free survival evaluated by BICR in accordance with RECIST v1.1 (BICR-PFS) in the Intent-to-Treat Population (Related to Figure 2).

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of PFS events	132 (51.4)	164 (63.8)
Disease progression	117 (45.5)	152 (59.1)
Death	15 (5.8)	12 (4.7)
Number of censored patients	125 (48.6)	93 (36.2)
Without death and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	1 (0.4)
PD or death after missing ≥2 consecutive tumor evaluations	6 (2.3)	16 (6.2)
PD or death after subsequent anti-tumor treatment	2 (0.8)	4 (1.6)
No PD, no death, and started subsequent anti-tumor treatment	19 (7.4)	10 (3.9)
No PD, no death, and no subsequent anti-tumor treatment	94 (36.6)	59 (23.0)
Progression-free survival (months)		
25% quantile (95% CI)	4.2 (4.0, 5.4)	3.1 (2.7, 4.1)
Median (95% CI)	5.7 (5.6, 7.0)	5.5 (5.2, 5.6)
75% quantile (95% CI)	12.4 (9.7, NE)	7.0 (5.8, 7.5)
Minimum, maximum	0.033+, 21.355+	0.033+, 19.285+
12-month PFS rate (%) (95% CI)	27.8 (20.4, 35.8)	6.1 (2.2, 12.6)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	NE (NE, NE)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.461, 0.738)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.451, 0.719)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: BIRC assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by BIRC per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 6. Progression-free survival evaluated by BICR in accordance with RECIST v1.1 (BICR-PFS) in the Per-protocol Population (Related to Figure 2)

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of PFS events	130 (51.0)	162 (63.5)
Disease progression	116 (45.5)	151 (59.2)
Death	14 (5.5)	11 (4.3)
Number of censored patients	125 (48.6)	93 (36.2)
Without death and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	1 (0.4)
PD or death after missing ≥2 consecutive tumor evaluations	6 (2.3)	16 (6.2)
PD or death after subsequent anti-tumor treatment	2 (0.8)	4 (1.6)
No PD, no death, and starting subsequent anti-tumor treatment	19 (7.4)	10 (3.9)
No PD, no death, and no subsequent anti-tumor treatment	94 (36.6)	59 (23.0)
Progression-free survival (months)		
25% quantile (95% CI)	4.2 (4.0, 5.4)	3.7 (2.7, 4.1)
Median (95% CI)	5.7 (5.6, 7.0)	5.5 (5.3, 5.6)
75% quantile (95% CI)	13.1 (9.7, NE)	7.0 (5.8, 7.5)
Minimum, maximum	0.033+, 21.355+	0.033+, 19.285+
12-month PFS rate (%) (95% CI)	28.3 (20.8, 36.3)	6.1 (2.3, 12.7)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	NE (NE, NE)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.59 (0.463, 0.744)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.453, 0.724)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: BIRC assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by BIRC per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 7. Progression-free survival evaluated by the investigator in accordance with RECIST 1.1 (INV-PFS) in the Intent-to-Treat Population (Related to Figure 2)

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of PFS events	128 (49.8)	167 (65.0)
Disease progression	109 (42.4)	149 (58.0)
Death between two tumor evaluations	19 (7.4)	18 (7.0)
Number of censored patients	129 (50.2)	90 (35.0)
Without death, and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	0
PD or death after missing ≥2 consecutive tumor evaluations	7 (2.7)	9 (3.5)
PD or death after subsequent anti-tumor treatment	0	1 (0.4)
No PD, no death, and starting subsequent anti-tumor treatment	14 (5.4)	5 (1.9)
No PD, no death, and no subsequent anti-tumor treatment	104 (40.5)	72 (28.0)
Progression-free survival (months)		
25% quantile (95% CI)	5.0 (4.2, 5.5)	3.4 (2.8, 4.1)
Median (95% CI)	7.0 (5.7, 8.1)	5.6 (5.5, 5.8)
75% quantile (95% CI)	13.1 (9.7, 19.5)	7.5 (7.0, 8.4)
Minimum, maximum	0.033+, 19.548+	0.033+, 19.318
12-month PFS rate (%) (95% CI)	27.6 (19.9, 35.8)	6.1 (2.4, 12.1)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	0 (0, 0)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.457, 0.735)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.449, 0.718)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: Investigator assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by the investigator per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 8. Progression-free survival evaluated by the investigator in accordance with RECIST v1.1 (INV-PFS) in the Per-protocol Population (Related to Figure 2)

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of PFS events	127 (49.8)	165 (64.7)
Disease progression	109 (42.7)	147 (57.6)
Death between two tumor evaluations	18 (7.0)	18 (7.0)
Number of censored patients	129 (50.2)	90 (35.0)
Without death, and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	0
PD or death after missing ≥2 consecutive tumor evaluations	7 (2.7)	9 (3.5)
PD or death after subsequent anti-tumor treatment	0	1 (0.4)
No PD, no death, and starting subsequent anti-tumor treatment	14 (5.4)	5 (1.9)
No PD, no death, and no subsequent anti-tumor treatment	104 (40.5)	72 (28.0)
Progression-free survival (months)		
25% quantile (95% CI)	5.0 (4.2, 5.5)	3.7 (2.8, 4.3)
Median (95% CI)	6.8 (5.7, 8.1)	5.6 (5.5, 5.8)
75% quantile (95% CI)	13.1 (10.6, 19.5)	7.5 (7.0, 8.4)
Minimum, maximum	0.033+, 19.548+	0.033+, 19.318
12-month PFS rate (%) (95% CI)	27.9 (20.2, 36.2)	6.1 (2.5, 12.2)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	0 (0, 0)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.59 (0.464, 0.749)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.58 (0.455, 0.730)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: Investigator assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by the investigator per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 9. Overall survival in the Intent-to-Treat population (Related to Figure 3).

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of deaths	70 (27.2)	103 (40.1)
Number of censored patients	187 (72.8)	154 (59.9)
Overall survival (months)		
25% quantile (95% CI)	8.1 (7.3, 10.9)	7.4 (6.9, 8.2)
Median (95% CI)	17.0 (14.0, NE)	11.0 (10.4, 12.6)
75% quantile (95% CI)	NE (NE, NE)	18.3 (13.4, NE)
Minimum, maximum	0.329, 23.918+	0.164, 24.641+
12-month survival rate (%) (95% CI)	66.0 (57.5, 73.2)	43.7 (34.4, 52.6)
24-month survival rate (%) (95% CI)	NE (NE, NE)	17.5 (8.7, 28.9)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.425, 0.783)	
p-value	0.0004	
Non-stratified analysis		
Hazard ratio (95% CI)	0.59 (0.433, 0.797)	
p-value	0.0006	

CI, confidence interval, NE, not estimable.

Note: OS is defined as the time from randomization to death from any cause and is calculated as (death date or censoring date - randomization date + 1)/30.4375 in month.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 10. Overall survival in the Per-protocol population (Related to Figure 3).

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of deaths	69 (27.1)	101 (39.6)
Number of censored patients	186 (72.9)	154 (60.4)
Overall survival (months)		
25% quantile (95% CI)	8.1 (7.3, 10.9)	7.5 (6.9, 8.2)
Median (95% CI)	17.0 (14.0, NE)	11.1 (10.4, 12.6)
75% quantile (95% CI)	NE (NE, NE)	18.3 (13.4, NE)
Minimum, maximum	0.329, 23.918+	0.164, 24.641+
12-month survival rate (%) (95% CI)	66.5 (58.1, 73.7)	44.1 (34.7, 53.0)
24-month survival rate (%) (95% CI)	NE (NE, NE)	17.7 (8.7, 29.2)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.429, 0.795)	
p-value	0.0006	
Non-stratified analysis		
Hazard ratio (95% CI)	0.59 (0.435, 0.805)	
p-value	0.0008	

CI, confidence interval, NE, not estimable.

Note: OS is defined as the time from randomization to death from any cause and is calculated as (death date or censoring date - randomization date + 1)/30.4375 in month.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 11. Objective response rate (INV-ORR), disease control rate (INV-DCR) and duration of response (INV-DoR) evaluated by investigators in accordance with RECIST v1.1 in the Intent-to-Treat Population (Related to Table 2).

Variable	Toripalimab + TP (N=257)	Placebo + TP (N=257)
Best overall response - no. (%)		
Complete response	13 (5.1)	8 (3.1)
Partial response	173 (67.3)	142 (55.3)
Stable disease	52 (20.2)	74 (28.8)
Progressive disease	10 (3.9)	24 (9.3)
Not evaluable	9 (3.5)	9 (3.5)
Objective response rate (ORR)		
ORR % (95% CI)	72.4 (66.5-77.7)	58.4 (52.1-64.5)
Difference in ORR % (95% CI)	14.1 (6.0-22.2)	
P-value[1]	0.0007	
Disease control rate (DCR)		
DCR % (95% CI)	92.6 (88.7-95.5)	87.2 (82.4-91.0)
Difference in DCR % (95% CI)	5.5 (0.3-10.7)	
P-value[1]	0.0371	
Estimate of median DoR (95% CI) (months)	7.0 (5.7-8.4)	4.3 (4.2-5.0)
Stratified estimate of hazard ratio (toripalimab versus placebo) (95%CI) [2]	0.50 (0.36-0.69)	
P-value[3]	0.0001	

TP, paclitaxel and cisplatin; CR, complete response; PD, progressive disease; NE, not estimable.

[1]In the stratified analysis, nominal p-value (two-sided) of the comparison between arms was calculated using Cochran-Mantel-Haenszel method based on random stratification factors, and the 95% confidence interval for the difference in the rate difference between the arms was estimated based on Mantel-Haenszel, stratified by ECOG PS score (0 vs. 1), and prior radiotherapy (yes vs. no).

[2]HR for DoR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

[3]P-value (two-sided) for DoR was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 12. Study Drug Exposure (Related to Table 3).

	Toripalimab +TP (N=257)	Placebo +TP (N=257)
The number of toripalimab/placebo cycles		
Mean	8.3	7.1
Sd	5.7	4.1
Median	7.0	7.0
Minimum, Maximum	1, 33	1, 26
Weeks of exposure for toripalimab/placebo		
Mean	23.8	19.7
Sd	17.8	13.0
Median	21.0	19.4
Minimum, Maximum	0.1, 96.1	0.1, 84.3
Cumulative dose for toripalimab/placebo (mg)		
Mean	1993	1709
Sd	1366	978
Median	1680	1680
Minimum, Maximum	240, 7920	240, 6240
Relative dose intensity for toripalimab/placebo (%)		
Mean	92.6	94.2
Sd	9.2	7.6
Median	95.6	96.6
Minimum, Maximum	36.5, 101.9	52.2, 102.5
The number of cycles cisplatin		
Mean	5.2	5.1
Sd	2.2	2.5
Median	6.0	6.0
Minimum, Maximum	1, 18	1, 18
Weeks of exposure for cisplatin		
Mean	12.9	12.2
Sd	5.4	5.5
Median	15.1	15.0
Minimum, Maximum	0.2, 27.9	0.2, 25.7
Cumulative dose for cisplatin (mg)		
Mean	569	554
Sd	207	207
Median	630	594
Minimum, Maximum	96, 876	80, 939

	Toripalimab +TP (N=257)	Placebo +TP (N=257)
Relative dose intensity for cisplatin (%)		
Mean	95.2	96.9
Sd	37.3	38.1
Median	92.7	94.2
Minimum, Maximum	36.1, 308.2	40.1, 359.5
The number of cycles paclitaxel		
Mean	5.0	4.8
Sd	1.5	1.6
Median	6.0	6.0
Minimum, Maximum	0, 6	1, 6
Weeks of exposure for paclitaxel		
Mean	13.3	12.5
Sd	5.1	5.3
Median	15.2	15.0
Minimum, Maximum	0.0, 27.9	0.1, 25.7
Cumulative dose for paclitaxel (mg)		
Mean	1382	1345
Sd	469	481
Median	1553	1471
Minimum, Maximum	0, 2043	28, 2192
Relative dose intensity for paclitaxel (%)		
Mean	88.6	90.8
Sd	11.4	11.3
Median	92.8	94.8
Minimum, Maximum	36.0, 109.8	9.1, 105.0

Sd, standard deviation.

Note: Actual dose intensity (ADI) (mg/week） = Average Actual Cumulative Dose received per week.

Note: Relative dose intensity (RDI) (%) = (Actual dose intensity (ADI) / Planned dose intensity) *100%, and Planned dose intensity is defined as first (first treatment cycle) planned dose divided by the number of weeks of 1 planned treatment cycle (e.g. 3 weeks).

Supplemental Table 13. Summary of Treatment Emergent Adverse Events (Safety Population. Related to Table 3)

Adverse Event Category	**Toripalimab + TP** (N=257) n (%)	**Placebo + TP** (N=257) n (%)
Any TEAE	255 (99.2)	255 (99.2)
Any TEAE related to study treatment	250 (97.3)	250 (97.3)
Any TEAE related to toripalimab/placebo	183 (71.2)	158 (61.5)
Any TEAE Grade ≥3	188 (73.2)	180 (70.0)
Any TEAE Grade ≥3 related to study treatment	166 (64.6)	144 (56.0)
Any TEAE Grade ≥3 related to toripalimab/placebo	61 (23.7)	34 (13.2)
Any TEAEs resulting in death	21 (8.2)	21 (8.2)
Any TEAEs resulting in death related to study treatment	1 (0.4)	3 (1.2)
Any TEAEs resulting in death related to toripalimab/placebo	1 (0.4)	2 (0.8)
Any SAE	93 (36.2)	74 (28.8)
Any SAE related to study treatment	60 (23.3)	38 (14.8)
Any SAE related to toripalimab/placebo	37 (14.4)	14 (5.4)
Any TEAE leading to discontinuation of toripalimab/placebo	30 (11.7)	16 (6.2)
Any TEAE leading to interruption of toripalimab/placebo	92 (35.8)	66 (25.7)
Any irAE [1]	95 (37.0)	68 (26.5)
Any irAE with Grade ≥3	18 (7.0)	4 (1.6)
Any IRR	9 (3.5)	8 (3.1)

TEAE: treatment-emergent adverse event, SAE: serious adverse event, irAE: immune-related adverse event, IRR: infusion-related reaction.

Note: TEAEs are adverse events that either start or worsen in severity on or after the date/time of first dose of study treatment and on or before 60 days after the date/time of last dose of study treatment or before starting a new anti-cancer therapy, whichever occurs first.

[1] As assessed by the Investigator

Data Cutoff: March 22, 2021

Supplemental Table 14. Immune related adverse events (irAE) reported in more than one patient in either arm as assessed by the investigator (Related to Table 3).

No. of patients (%)	Toripalimab + TP (N=257)		Placebo + TP (N=257)	
	All Grades	Grades ≥3	All Grades	Grade ≥3
Any irAE	95 (37.0%)	18 (7.0%)	68 (26.5%)	4 (1.6%)
Rash	26 (10.1%)	5 (1.9%)	9 (3.5%)	0
Hypothyroidism	15 (5.8%)	0	9 (3.5%)	0
Pruritus	11 (4.3%)	1 (0.4%)	4 (1.6%)	0
Fatigue	10 (3.9%)	1 (0.4%)	9 (3.5%)	0
Hyperglycemia	10 (3.9%)	0	7 (2.7%)	0
Thyroid function test abnormal	10 (3.9%)	0	5 (1.9%)	0
Alanine aminotransferase increased	7 (2.7%)	1 (0.4%)	4 (1.6%)	0
Aspartate aminotransferase increased	6 (2.3%)	1 (0.4%)	3 (1.2%)	0
Diarrhea	6 (2.3%)	0	2 (0.8%)	0
Neutropenia	5 (1.9%)	1 (0.4%)	3 (1.2%)	0
Pneumonitis	5 (1.9%)	0	2 (0.8%)	0
Anemia	4 (1.6%)	0	6 (2.3%)	1 (0.4%)
Arrhythmia	4 (1.6%)	0	3 (1.2%)	0
Decreased appetite	4 (1.6%)	0	4 (1.6%)	0
Hepatic function abnormal	4 (1.6%)	1 (0.4%)	0	0
Leukopenia	4 (1.6%)	0	1 (0.4%)	0
Nausea	4 (1.6%)	0	7 (2.7%)	0
Blood creatine phosphokinase increased	3 (1.2%)	0	0	0
Constipation	3 (1.2%)	0	3 (1.2%)	0
Hyperthyroidism	3 (1.2%)	0	0	0
Thrombocytopenia	3 (1.2%)	0	1 (0.4%)	1 (0.4%)
Vomiting	3 (1.2%)	0	3 (1.2%)	0
Blood urea increased	2 (0.8%)	0	0	0
Enteritis	2 (0.8%)	2 (0.8%)	0	0
Gamma-glutamyl transferase increased	2 (0.8%)	1 (0.4%)	0	0
Heart injury	3 (1.2%)	1 (0.4%)	1 (0.4%)	0
Hypertension	2 (0.8%)	1 (0.4%)	0	0
Immune-mediated myocarditis	1 (0.4%)	1 (0.4%)	0	0
Pyrexia	2 (0.8%)	0	2 (0.8%)	0
Immune-mediated hepatitis	1 (0.4%)	1 (0.4%)	2 (0.8%)	1 (0.4%)
Hyperlipidemia	1 (0.4%)	0	2 (0.8%)	0
Musculoskeletal pain	1 (0.4%)	0	4 (1.6%)	0
Arthritis	0	0	2 (0.8%)	0
Myelosuppression	0	0	3 (1.2%)	1 (0.4%)
Proteinuria	0	0	2 (0.8%)	0

Note: For the same patient if the same adverse event occurs multiple times, that patient is counted only once in the same system organ classification and preferred terminology when calculating the number of adverse events.
Data Cutoff: Mar 22 2021

Supplemental Table 15. Comparability study among different PD-L1 IHC staining antibodies including JS311, 22C3 and 28-8 in tumor biopsies from ESCC patients (n=50). JS311 (CPS≥10) showed 84% and 90% overall concordance with 22C3 (CPS≥10) and 28-8 (TPS≥1) for PD-L1 IHC staining in ESCC (Related to STAR Methods).

ESCC (n=50)		**22C3（CPS≥10）**		**28-8(TPS≥1%)**	
		+	-	+	-
JS311（CPS≥10）	+	1	8	6	3
	-	0	41	2	39